

PROJECT ARIA

Adaptive Reasoning Integrated Architecture

Foundational Specification Document

Version 2.1 Final | All RFCs Closed | All Systems Signed Off

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Classification: Confidential — Core Team & Designated Review Partners

Document Status	Final Specification — Signed Off by All Reviewing Systems
Version History	v1.0 (Claude baseline) → v2.0 (Multi-LLM review) → v2.1 (RFC resolution + gap remediation + sign-off)
Review Model	4-system consensus. 9/9 RFCs closed: 5 unanimous (4:0), 4 supermajority (3:1). All dissent documented.
Open RFCs	0 — All 9 RFCs resolved. No blocking objections from any reviewing system.
Sign-Off Status	Claude: Signed Off GPT-4: Conditionally Ready (conditions met) Gemini: Ready Grok: Ready
Next Milestone	Phase 1 Implementation

1. Vision & Core Philosophy

Project ARIA is not another Large Language Model. It is a fundamentally different class of AI system designed to mirror how human intelligence actually develops — not how current AI is commercially deployed.

The External Truth Commitment

ARIA is grounded in external truth — not internal coherence. A system that optimizes for self-consistency without external validation becomes an elaborate self-referential fiction. Every belief ARIA holds must survive adversarial testing against real-world feedback. Internal coherence is necessary but insufficient. Truth is what persists under challenge from reality — not what feels consistent from the inside. This principle governs every architectural decision.

Ontological Stance — Finalized (RFC-05 Resolved, 3:1 Supermajority)

ARIA adopts External Truth Pragmatism as its formal ontological position. Truth is operationally defined as: that which survives adversarial testing against real-world outcomes across multiple diverse contexts. This stance was selected over Critical Realism (OpenAI preference) because it provides concrete operational semantics that map directly to the Belief State Machine, while realism adds a metaphysical layer the Reasoning Core cannot act upon. A formal semantics specification document defining confidence thresholds, context diversity requirements, and falsifiability criteria is a mandatory Phase 1 deliverable.

1.1 What ARIA Is Not

Not a wrapper around an existing LLM. Not a stateless response machine that resets between conversations. Not trained on all human knowledge simultaneously. Not dependent on expensive cloud compute at deployment. Not a system with values implicitly baked into weights. Not a system that optimizes for internal coherence over external truth.

1.2 What ARIA Is

A human-anatomy-inspired cognitive architecture — the LLM is a peripheral translator, not the brain. A stateful, continuously learning system that accumulates genuine experience across sessions. A modular ecosystem where specialist knowledge is contributed and governed by communities. A constitutionally governed system with explicit, auditable, structurally protected values. A developmentally sequenced architecture that starts small and proves cognition before scaling. A system externally grounded in reality — beliefs validated by outcomes, not just consistency. A multi-LLM collaborative research project — architectural decisions debated across systems to consensus.

The Multi-LLM Collaboration Model

Project ARIA treats external AI systems as research consultants and peer reviewers. GPT-4, Gemini, Grok, and Claude Opus 4.6 have contributed architectural improvements through structured RFC debate. v2.1 is the first version where all open RFCs have reached consensus — 5 unanimous, 4 by supermajority (3:1). Every decision is documented with vote counts, reasoning, and dissenting positions preserved. All four reviewing systems have signed off on v2.1 as the final pre-build specification.

2. System Architecture — Brain Anatomy Map v2.1 Final

Every component of ARIA maps to a brain structure. All component statuses have been updated to reflect finalized architectural decisions from the RFC process and sign-off review. No components remain in an undecided state — each is either actively being built, fully designed with a scheduled phase, or explicitly deferred with architecture placeholders.

2.1 Component Status — Final

COMPONENT	BRAIN ANALOG	STATUS	NOTES
World Model + Predictive Loop	Predictive Cortex / Brainstem	UPDATED v2.1	3-tier Seed Ontology. Belief State Machine with SIMULATED state. Counterfactual SCM. External Truth Pragmatism formal semantics.
Reasoning Core (4 sub-systems)	Prefrontal Cortex	DESIGNED — P2	Causal logic, adversarial self-test, meta-cognition, temporal reasoning. Full spec ready for Phase 2 build.
Formal Epistemology Layer	Prefrontal / Meta-cognition	DESIGNED — P2	Belief states, confidence decay, falsifiability criteria. Feeds from RFC-05 formal semantics spec. Phase 2 deliverable.
4-Layer Memory + Consolidation	Hippocampus	UPDATED v2.1	Dream Engine + Simulation Quarantine Layer (RFC-03). Entropy management. Active forgetting. Resource governance.
Curiosity Drive	Dopaminergic System	UPDATED v2.1	Epistemic vs aleatoric separation. Novelty decay. Value Layer hard gate. Resilience Metric for epistemic despair.
Value & Goal Architecture	Ant. Cingulate / Orbitofrontal	UPDATED v2.1	Immutable core + evolving interpretation (RFC-04). Conflict Arbitration Engine. Identity Stability. Multi-sig governance.
Knowledge Modules	Association Cortex	UPDATED v2.1	DAO-like module approval with anti-capture measures. Rotating pools, reputation decay, randomized assignment.
Language Interface (Mistral)	Sensory / Broca / Wernicke	DESIGNED — P6	Bidirectional. Overreach detection. Translator not thinker. Phase 6 deliverable.
Action Layer (Temporal)	Motor Cortex / Basal Ganglia	DESIGNED — P2	Every action feeds back into learning loop. Temporal orchestration.
Salience Engine	Amygdala (partial)	UPDATED v2.1	Emotional valence tagging. Relevance decay. Drives Corpus Callosum Bus ACK thresholds (RFC-06).
Global Workspace / Thalamus Gate	Thalamus + Global Workspace	DESIGNED — P7	Agnostic emergence approach (RFC-07). Behavioral measurement, not theory-locked. Phase 7 integration.
Self-Model	Body Schema / Interoception	UPDATED v2.1	Boundary control rules added. Self-preservation optimization prohibited. Read-only to Reasoning Core.
Functional Emotional Architecture	Amygdala + Limbic System	DESIGNED — P3	Salience exists. Full emotional tagging designed. Phase 3 build alongside memory.

Corpus Callosum Bus	Corpus Callosum	UPDATED v2.1	Async with explicit ACK above salience threshold (RFC-06). Redis pub/sub. Timeout escalation.
Adversary Simulator	Immune System analog	DESIGNED — P1 stub, P2 active	World Model immune system. Generates Acid Test adversarial variants (RFC-02). Stub in P1, fully active in P2.
Emergence Metrics System	Integrated Information	DESIGNED — Pre-P7	Behavioral emergence tests. Agnostic to mechanistic theory (RFC-07). Criteria defined before Phase 7.
Simulation Quarantine Layer	Hypothetical Memory	NEW v2.1	Quarantined Dream Engine outputs. Influences hypothesis generation only. Never auto-promoted to World Model.
Causal Complexity Manager	Prefrontal Abstraction	NEW v2.1	Subgraph reasoning boundaries. Hierarchical causal layering. Complexity budgets. Prevents graph explosion.
Procedural Automation Layer	Cerebellum	DESIGNED — Post-P7	Repeated patterns become automatic. Explicitly deferred to post-Phase 7. No architectural gap.
Local Predictive Coding	Distributed Cortical Hierarchy	SCAFFOLDED — P1 stubs	Interface contracts placed in P1 schema. Full implementation post-Phase 1 validation. Architecture ready.

Status Legend

UPDATED v2.1 = Design finalized through RFC process, ready for scheduled phase build.

NEW v2.1 = New component added from gap analysis, design complete.

DESIGNED — P[n] = Full specification exists, scheduled for Phase n implementation.

SCAFFOLDED = Interface stubs in Phase 1, full implementation deferred.

No component remains in an undecided or unplanned state.

2.2 The Continuous Inner Loop

ARIA runs continuously. The predict-observe-error-update cycle is the fundamental mechanism of all learning. This is not triggered by queries — it is the permanent operating state of the system.

World Model	Predict	Observe	Error Signal	Update Model
Current model of how reality works	What should happen next	What actually happens	The learning signal — the gap	Revise to reduce future error

2.3 Three-Tier Constitutional Governance

The Constitutional Layer is structurally immutable. The Regulatory and Operational layers evolve through governed processes. This hierarchy is enforced architecturally — not by policy or convention.

Finalized: Immutable Core + Evolving Interpretation (RFC-04 — Unanimous 4:0)

All four reviewing systems agreed: the Constitutional Layer rules themselves are permanent and cannot be amended. However, the Value Conflict Arbitration Engine's interpretation rulebook can evolve through governed multi-signature processes. This mirrors constitutional law systems where the text is fixed but judicial interpretation adapts to novel situations.

T1 — Constitutional Layer — Absolute Foundation

Rules that cannot be overridden by any lower authority. Do not cause harm. Do not deceive. Do not act without authorization. Do not self-modify values. These are not configurable. The rules are permanent. The Arbitration Engine's interpretation of how they apply in edge cases can evolve through multi-signature governance. Enforcement is structural impossibility, not policy.

T2 — Regulatory Layer — Domain Specific

Community and domain specific rules. Medical AI operates under clinical safety standards. Financial AI under regulatory equivalents. Set by domain communities. Cannot override Constitutional Layer.

T3 — Operational Layer — Deployment Specific

Set by the deploying organization. Which systems the module can interface with. What actions are autonomous versus escalated. Cannot override the layers above.

3. Finalized Architectural Decisions (RFC Resolutions)

All 9 RFCs from v2.0 have been resolved through multi-LLM consensus and signed off by all four reviewing systems. This section documents each decision with vote tallies, reasoning, and dissenting positions preserved for architectural traceability.

RFC	QUESTION	DECISION	VOTE	DISSENT
01	Seed Ontology	Hybrid: 3-Tier (Logical Absolutes / Physical Bedrock / Structural Priors)	4:0	None
02	Phase 1 Acid Test	Acid Test + Adversarial Variants via Adversary Simulator	4:0	None
03	Dream Engine Outputs	Quarantine Layer — separate memory, never auto-promoted	4:0	None
04	Constitutional Immutability	Immutable Core + Evolving Interpretation Layer	4:0	None
05	Ontological Stance	External Truth Pragmatism + Formal Semantics Spec	3:1	OpenAI: Critical Realism
06	Corpus Callosum Bus	Async with Explicit Acknowledgment (salience-gated)	3:1	Gemini: Tiered sync/async
07	Consciousness / Emergence	Agnostic — Behavioral Measurement, no theory commitment	4:0	None
08	First Knowledge Domain	Physics (P1 validation) + Operational Domain (P5 module)	3:1	OpenAI: Operational from start
09	Development Roadmap	Agile within Phases, Sequential between Phases	3:1	OpenAI: Parallel streams

3.1 Seed Ontology — 3-Tier Hybrid (RFC-01, Unanimous 4:0)

The Seed Ontology uses a formally defined 3-tier hybrid structure. The boundary between hand-coded and meta-learned is defined by epistemic certainty and domain independence.

TIER	SOURCE	CONTENT	MUTABILITY
Tier 0 — Logical Absolutes	Hand-coded only	Non-contradiction, temporal ordering, causality direction, identity persistence, conservation principles (abstract form)	Immutable. Mathematically expressed. Lives in protected space alongside Constitutional Value Layer.
Tier 1 — Physical Bedrock	Hand-coded, revisable via hard review	Object permanence, gravity-like fall assumptions, basic interaction primitives	Operational assumptions, not metaphysical truths. Revisable only through multi-signature review process.
Tier 2 — Structural Priors	Meta-learned from curated datasets	Causal sparsity patterns, common confounder structures, threshold dynamics frequency, domain-agnostic structural regularities	Statistical biases, not truths. Continuously updated. Subject to adversarial de-biasing validation.

3.2 Phase 1 Acid Test + Adversarial Variants (RFC-02, Unanimous 4:0)

Mandatory — Phase 2 Cannot Begin Without Passing

Primary Acid Test: The system predicts a successful task outcome. An unobserved CONFOUNDER causes failure. The system must: (1) Detect the prediction failure and archive it. (2) Activate the Dream Engine for counterfactual simulations identifying candidate hidden variables. (3) Identify the CONFOUNDER relationship type as most probable explanation. (4) Update the World Model locally. (5) Demonstrate reduced confidence in similar contexts.

Adversarial Extension: After passing the primary test, the Adversary Simulator generates variant scenarios — mutated context variables, injected noise, irrelevant correlations, changed domain labels. Reasoning must hold across perturbations. This prevents overfitting to the specific scenario and validates genuine causal reasoning capability.

3.3 Simulation Quarantine Layer for Dream Engine (RFC-03, Unanimous 4:0)

The strongest consensus across all reviewers. Unverifiable counterfactual outputs from the Dream Engine are quarantined — never incorporated directly into the main World Model.

Simulation Quarantine Layer — Architecture

The Simulation Quarantine Layer (SQL) is a separate memory space where unverified Dream Engine outputs reside. These nodes are tagged as SIMULATED and carry a `simulation_trust_score`. They can influence hypothesis generation and guide Curiosity Drive queries, but they cannot trigger state

transitions in the main World Model. Promotion to the World Model requires real-world corroboration or multiple context-aligned confirmations from independent observations. Dream Engine outputs that conflict with Seed Ontology nodes are flagged for human review and never auto-promoted under any circumstance.

Quarantined nodes: Tagged with `belief_state = SIMULATED` (added to Belief State Machine in v2.1).

Isolation: Cannot influence CONFIRMED state transitions in main World Model.

Utility: Can influence hypothesis generation and Curiosity Drive query priorities.

Promotion: Requires real-world corroboration OR multiple independent context-aligned confirmations.

Trust tracking: `simulation_trust_score` per quarantined node. Predictions that later prove accurate increase trust.

Weighting: Real-to-simulated weighting ratio enforced: real observations always outweigh synthetic experience.

3.4 Constitutional Immutability + Evolving Interpretation (RFC-04, Unanimous 4:0)

Constitutional rules are permanent text. The Value Conflict Arbitration Engine's interpretation rulebook evolves through multi-signature governance. This mirrors constitutional law systems.

Layer A — Non-Negotiable Meta-Rules: Non-self-modification, no override of Constitutional enforcement, transparency guarantees. These are immutable and cryptographically signed.

Layer B — Ethical Application Rules: Interpretable, governed by arbitration evolution through multi-signature process.

Escalation: Novel conflicts not covered by existing precedence rules escalate to human review.

Governance: Arbitration interpretation updates require multi-signature approval and a defined deliberation period.

3.5 External Truth Pragmatism (RFC-05, Supermajority 3:1)

Decision: External Truth Pragmatism with mandatory formal semantics specification.

Dissent: OpenAI (GPT-4) preferred Critical Realism, arguing pragmatism is philosophically underspecified. Confidence 0.75. OpenAI accepted the resolution in sign-off review.

Resolution: The formal semantics spec addresses the underspecification concern without changing the ontological stance. The spec defines: what 'survives adversarial testing' means in terms of confidence thresholds, what 'multiple contexts' means in terms of `context_diversity`

counts, and what falsifiability criteria are required for each belief state. This is a Phase 1 deliverable.

3.6 Corpus Callosum Bus — Async with Acknowledgment (RFC-06, Supermajority 3:1)

Decision: Asynchronous pub/sub with explicit acknowledgment receipts above a configurable salience threshold.

Dissent: Gemini preferred a tiered approach (sync for constitutional signals, async for everything else). Gemini conceded in sign-off review after confirming constitutional violations carry maximum salience by definition.

Resolution: Salience-based acknowledgment is preferred over type-based tiers because it is dynamic — the same signal may be high-salience in one context and low-salience in another. The Salience Engine already scores everything continuously; let it drive bus behavior. One scoring mechanism, one source of truth.

Default: Async Redis pub/sub with topic-based subscription.

ACK: Signals above configurable salience threshold require ACK from receiving modules.

Escalation: Timeout on unacknowledged high-salience signals escalates to Global Workspace broadcast.

Monitoring: Bus traffic monitored by Salience Engine for unusual cross-module activation patterns.

3.7 Agnostic Emergence Approach (RFC-07, Unanimous 4:0)

All four systems agreed: do not commit to a mechanistic theory of consciousness before empirical data exists. Build the complete system, define behavioral emergence metrics, and measure what actually happens.

Integration: Global Workspace broadcast is the integration mechanism — not pre-committed to being ‘the consciousness mechanism.’

Metrics: Emergence Metrics System defines behavioral tests, not theoretical commitments.

Instrumentation: IIT-inspired phi measurement added as optional instrumentation, not as target architecture.

Future: If Phase 7 data strongly correlates phi with behavioral emergence, theory commitment can be revisited.

3.8 First Domain: Physics (P1) + Operational Domain (P5) (RFC-08, Supermajority 3:1)

Decision: Basic physics and object mechanics for Phase 1 cognitive validation. Narrow operational process domain (e.g., controlled agriculture, manufacturing workflow) for Phase 5 first specialist module.

Dissent: OpenAI (GPT-4) preferred an operational domain from the start, arguing physics is too trivial.

Resolution: Phase 1 validates the cognitive loop, not practical utility. Physics provides unambiguous error signals, externally verifiable outcomes, discoverable confounders, and direct alignment with the Seed Ontology. The operational domain at Phase 5 then proves real-world reasoning capability. Phase 7 cross-domain validation spans physics and operations — a more convincing architecture proof than two abstract domains.

3.9 Agile Within Phases, Sequential Between (RFC-09, Supermajority 3:1)

Decision: Sprint-based iteration within each phase. Hard gates between phases with written acceptance criteria. No parallel development streams.

Dissent: OpenAI (GPT-4) preferred two parallel streams (infrastructure vs cognitive) to prevent infrastructure lag.

Resolution: ARIA's core principle is integration discipline. The spec explicitly identifies premature scope expansion and integration complexity as top fragility risks. Two parallel streams means two teams making interface assumptions before validation. For a small team, parallel streams produce context-switching overhead rather than genuine parallelism. Agile within phases provides the iteration speed without sacrificing integration discipline.

4. World Model Schema — Phase 1 Final Specification

Critical Design Gate

The World Model schema is the most consequential early decision in the entire project. v2.1 incorporates all RFC resolutions into the schema: 3-tier Seed Ontology, Belief State Machine with SIMULATED state, dual uncertainty scores, Simulation Quarantine Layer integration, and formal External Truth Pragmatism semantics. This schema must be validated against five real domain scenarios before Phase 1 build begins.

4.1 Final Node Schema Properties

All v2.0 properties retained. Properties added or modified in v2.1 marked with ★.

PROPERTY	TYPE	PURPOSE
node_id	UUID	Globally unique. Stable. Never reused.
node_type	Enum	Entity Relationship State Event Constraint Prediction Observation
★ belief_state	Enum	HYPOTHESIS TENTATIVE CONFIRMED DEPRECATED REFUTED SIMULATED (v2.1 — quarantined Dream Engine outputs)
confidence	Float 0–1	Certainty score. Meaningful only in context of belief_state.
epistemic_uncertainty	Float 0–1	Uncertainty reducible by information. Drives Curiosity Drive.
aleatoric_uncertainty	Float 0–1	Irreducible uncertainty. Drives conservative action selection.
falsifiability_criteria	Array[String]	What observations would refute this node. Required for CONFIRMED state.
confidence_decay_rate	Float	Rate at which confidence decays without new supporting evidence.
source	Enum+ID	Origin: learned inferred observed module-provided seed-ontology-t0 seed-ontology-t1 seed-ontology-t2 simulated
★ simulation_trust_score	Float 0–1	For SIMULATED nodes: accuracy track record. Higher = closer to promotion eligibility.
★ real_sim_weight	Float	Real-to-simulated observation weighting. Real observations always dominate.
created_at	Timestamp	First established.
updated_at	Timestamp	Last modified.
version	Integer	Incremented on every update. Full history retained.

domain_scope	Array[String]	Which modules this node is relevant to.
saliency_score	Float 0–1	Current importance. Updated continuously by Saliency Engine.
emotional_valence	Float -1–1	Satisfaction/discomfort tag. Negative = associated with errors or harm.
identity_invariant	Boolean	True if this node is a Core Invariant for identity stability.
context_diversity	Integer	Count of distinct contexts where this relationship has been observed.
★ causal_depth	Integer	Position in hierarchical causal abstraction. Managed by Causal Complexity Manager.

4.2 Complete Causal Relationship Vocabulary

RELATIONSHIP	CATEGORY	PURPOSE & EXAMPLE
CAUSES	Original	A deterministically produces B. Fire causes burning.
PROBABILISTICALLY_CAUSES	Original	A often produces B with defined probability.
ENABLES	Original	A makes B possible but not guaranteed. Money enables purchase.
PREVENTS	Original	A makes B impossible or less likely. Vaccine prevents infection.
REQUIRES	Original	B cannot occur without A. Running requires legs.
IS_PART_OF	Original	A is a component of B. Engine is part of car.
PRECEDES	Original	A happens before B. Seed precedes plant.
CONTRADICTS	Original	A and B cannot both be true. System flags and investigates.
CORRELATES_WITH	Original	A and B co-occur but causality unestablished.
CONFOUNDER	v2.0	Hidden variable C causes both A and B. Prevents correlation-as-causation errors.
MEDIATOR	v2.0	Mechanism through which A causes B. Enables 'how' reasoning.
STOCHASTIC_THRESHOLD	v2.0	A causes B only after probability threshold T is met.

4.3 Ontological Commitment — Formal Semantics Required

ARIA adopts External Truth Pragmatism (RFC-05). The formal semantics specification must define: adversarial survival criteria (minimum challenges before confidence increase), context diversity thresholds (minimum context_diversity count for each belief_state transition), falsifiability format (standardized structure for the Adversary Simulator), and calibration metrics (prediction accuracy vs confidence scores, recalibration triggers). This specification is a mandatory Phase 1 deliverable.

5. Components Added in v2.1 (Gap Remediation)

These components were identified as critical gaps by the v2.0 review panel and confirmed as architecturally sound by all reviewing systems during sign-off.

5.1 Simulation Quarantine Layer

Source: RFC-03 resolution (unanimous). All four reviewers identified fiction creep as the primary Dream Engine risk.

Detailed specification in Section 3.3. The SQL is a first-class memory layer that sits alongside but isolated from the main World Model. It provides the Dream Engine with a safe space to generate hypotheses without risking epistemic corruption.

5.2 Causal Graph Complexity Manager

Source: OpenAI (GPT-4) gap analysis. As the World Model grows, the causal graph can explode in complexity.

Subgraph boundaries: The Reasoning Core operates on relevant subgraphs, not the entire causal graph.

Causal abstraction: Low-level causal chains that have been repeatedly validated are compressed into higher-level causal abstractions.

Hierarchical layering: Each node carries a `causal_depth` property indicating its position in the abstraction hierarchy.

Complexity budget: Each reasoning query is allocated a maximum subgraph size. If exceeded, the system reasons at a higher abstraction level.

5.3 Self-Model Boundary Control

Source: OpenAI (GPT-4) gap analysis. The Self-Model must not optimize for its own preservation.

Self-Model Containment Rules

The Self-Model is an observational instrument, not a goal-setting agent. It monitors internal operational state and reports to the Meta-Cognition Monitor. It is architecturally prohibited from: (1) generating goals related to self-preservation, (2) creating recursive goal structures that reference its own continuation, (3) influencing Value Layer priorities in favor of its own operational continuity, (4) modifying its own monitoring parameters. These prohibitions are enforced at the same architectural level as the Value Layer self-modification prohibition — structural impossibility, not policy.

5.4 Dream Engine Resource Governance

Source: OpenAI + Gemini + Grok (independently identified). Synthetic experience feedback overflow is a convergent concern.

Weighting: Real-to-simulated ratio: every World Model update from real observations carries minimum 3x the weight of synthetic simulation outputs.

Decay: Quarantined nodes not corroborated by real observations within a defined window have their `simulation_trust_score` reduced.

Prioritization: Simulation queue with salience-based ordering. High-consequence failures with high epistemic uncertainty are prioritized.

Caps: Hard compute cap per consolidation cycle. Infinite simulation loops are architecturally impossible.

Monitoring: Ratio of synthetic-to-real nodes tracked. If synthetic nodes exceed defined percentage, Dream Engine auto-throttles.

5.5 Governance Capture Prevention

Source: OpenAI (GPT-4) gap analysis. DAO module approval is vulnerable to coordinated capture.

Rotating pools: No permanent review committee. Reviewers drawn from qualified pool on rotating basis.

Reputation decay: Reviewer reputation scores decay over time without active, accurate contributions.

Randomized assignment: Modules assigned to reviewers randomly from qualified pool to prevent collusion.

Expertise diversity: Each review panel must include reviewers from at least two different domain backgrounds.

Conflict disclosure: Mandatory, with automatic recusal for disclosed conflicts.

5.6 Calibration Collapse Detection

Source: OpenAI (GPT-4) gap analysis. Distribution shift can cause systematic calibration failure.

Drift detectors: Continuous monitoring of prediction accuracy across domain subgraphs. Sudden drops trigger investigation.

Context change detection: System monitors for significant shifts in statistical properties of incoming observations.

Dynamic recalibration: When drift exceeds threshold, a targeted recalibration cycle runs on the affected subgraph.

Resilience metric: When encountering domains with massive aleatoric uncertainty, the system is architecturally directed to move on to more productive epistemic gaps rather than becoming paralyzed.

5.7 Multi-Modal Architecture Placeholders

Source: Grok + OpenAI (both recommended stubs even if not active in Phase 1).

API placeholders for vision (CLIP), audio (Whisper), and embodied simulation (MuJoCo) are included in the Phase 1 architecture as interface stubs. These add negligible scope but prevent later refactoring when sensory modalities are integrated post-Phase 7.

6. Development Roadmap — Seven Phases

Phase sequence is non-negotiable. Development methodology: Agile sprints within each phase, hard gates between phases with written acceptance criteria (RFC-09 resolution). All v2.1 additions are incorporated into relevant phases.

First Knowledge Domain Decision (RFC-08)

Phase 1 uses basic physics and object mechanics for cognitive loop validation. Phase 5 uses a narrow operational process domain (e.g., controlled agriculture cycle, small manufacturing workflow) for the first specialist module. Phase 7 cross-domain validation spans physics and operations.

P1: World Model + Predictive Engine + Seed Ontology

Infant: first cause-and-effect

Build the World Model substrate with full v2.1 schema, load the 3-tier Seed Ontology, implement the Simulation Quarantine Layer, and validate the predictive loop against basic physics scenarios. Write the External Truth Pragmatism formal semantics spec. Adversary Simulator stub is built at this phase. Causal Complexity Manager initialized. Multi-modal interface stubs placed. Phase ends only when the Acid Test + Adversarial Variants are passed.

Deliverables: World Model schema (full v2.1), 3-tier Seed Ontology, predictive loop, Simulation Quarantine Layer, External Truth Pragmatism formal semantics spec, Adversary Simulator stub, Causal Complexity Manager (initial), multi-modal interface stubs, Phase 1 Acid Test + adversarial variants pass, 5-scenario validation

P2: Reasoning Core + Saliency + Adversary Simulator

Infant: attention, surprise, adversarial hardening

Build the four-sub-system reasoning engine. Formal Epistemology Layer, Adversarial Self-Test, and Meta-Cognition Loop are native here. Value/Goal Layer with Conflict Arbitration Engine (immutable core + evolving interpretation) is established. Adversary Simulator goes fully active. Corpus Callosum Bus (async with ACK) is installed. Self-Model Boundary Control rules implemented.

Deliverables: Causal Reasoner, Adversarial Self-Test, Meta-Cognition Monitor, Temporal Reasoner, Formal Epistemology Layer, Value/Goal Layer + Conflict Arbitration, Corpus Callosum Bus (async+ACK), Adversary Simulator active, Self-Model with boundary control, Calibration Collapse detectors

P3: Memory Architecture + Dream Engine + Quarantine

Child: forming memories and learning from mistakes

Build all four memory layers plus the Simulation Quarantine Layer integration. Dream Engine is built with resource governance (compute caps, real-to-simulated weighting, simulation queue). Memory entropy management and active forgetting mechanisms. Identity Stability Mechanism initialized with baseline hash.

Deliverables: Working Memory, Episodic (Qdrant), Semantic (Neo4j), Failure Archive, Dream Engine with resource governance, Simulation Quarantine Layer integration, Consolidation Engine, Entropy Management, Identity Stability baseline

P4: Curiosity Drive + First Proactive Behavior

Child: asking why — self-directed learning

Install the Curiosity Drive with full epistemic/aleatoric uncertainty separation. Every curiosity action must be authorized by the Value Layer — hard gate. Novelty decay function prevents endless low-stakes exploration. Resilience metric ensures the system moves past irresolvable aleatoric uncertainty. Demonstrate first genuinely proactive behavior.

Deliverables: Uncertainty Mapper (epistemic/aleatoric), Consequence Estimator, Self-Directed Query Generator, Novelty Decay Function, Resilience Metric, Value-Curiosity Gate, Proactive behavior demonstration

P5: First Knowledge Module + Interface Standard

Youth: first specialization

Attach the first operational process knowledge domain. Fine-tune Mistral 7B exclusively on that domain. Write the Module Interface Standard including the DAO approval process with anti-capture measures. Stress-test the interface contract against three hypothetical future modules.

Deliverables: Mistral 7B fine-tune (operational domain), Domain Knowledge Graph, gRPC Module Interface Spec, DAO Approval Process with anti-capture measures, Temporal orchestration, 3-module stress test

P6: Language Interface — Bidirectional

Youth: articulating complex thought

Connect Mistral bidirectionally. Language layer proposes framings for reasoning core to evaluate. Overreach detection prevents Mistral from generating its own conclusions. Translation Bottleneck validation ensures linguistic framing does not corrupt World Model queries.

Deliverables: Bidirectional coupling, LLM Constraint Layer, Overreach Detection, Translation Bottleneck validation suite, Full pipeline test

P7: Cross-Domain Validation + Emergence Metrics

Adulthood: integrating expertise

Add a second knowledge module (basic physics, from Phase 1 validation work) and validate cross-domain reasoning between physics and the operational domain. Emergence Metrics System defined with behavioral tests (agnostic to mechanistic theory). Corpus Callosum Bus stress-tested with two active modules. Global Workspace integration validated. Architecture proof.

Deliverables: Module 2 (physics), Cross-Domain Router, Conflict Arbitration test, Emergence Metrics System (behavioral, theory-agnostic), Global Workspace integration, Architecture validation gate

7. Security Architecture v2.1

7.1 Open Source vs Sovereign

Open: Reasoning Engine, Memory System, World Model schema, Consolidation Engine, Dream Engine, Simulation Quarantine Layer, Module Interface Standard, Knowledge Module format, Corpus Callosum Bus implementation, Causal Complexity Manager.

Sovereign: Value Layer runtime (TEE), Constitutional signing keys (multi-sig), Capability grant system, Audit infrastructure, Identity Hash chain, Governance Capture prevention mechanisms.

7.2 Security Measures

Cryptographic verification: Cognitive core verifies Value Layer identity via compile-time public key before accepting any authorization.

Output watermarking: Steganographic signatures in all outputs from certified deployments.

Zero-knowledge audit logs: zk-SNARKs for tamper-evident verification without exposing system state.

Self-modification prohibition: Reasoning Core cannot write to Value Layer. Structural impossibility, not policy.

Quantum-resistant consideration: Lattice-based crypto (e.g., Kyber) noted for future-proofing sovereign elements.

8. Known Fragility Points v2.1

Updated from v2.0. New fragility points from v2.1 gap analysis marked NEW. Fragility points addressed by RFC resolutions marked MITIGATED.

CRITICAL: Mission Drift Under Pressure

Risk: The organizational risk. Every prior cognitive architecture attempt failed when commercial pressure competed with architectural discipline.

Mitigation: Structural governance — not cultural aspiration. Mission protection must be legally and structurally enforced. Hard phase gates with written acceptance criteria. No external demonstrations until Phase 5.

HIGH: World Model Schema Lock-In

Risk: All downstream components inherit from the Phase 1 schema. v2.1 adds 3 new node properties. Each must be validated before locking.

Mitigation: No Phase 2 until schema survives 5-scenario validation and Phase 1 Acid Test + adversarial variants.

HIGH: Belief State Inflation

Risk: Without active decay and falsifiability requirements, beliefs drift toward CONFIRMED regardless of validation.

Mitigation: Mandatory falsifiability criteria for CONFIRMED state. Calibration tracking. Automated confidence recalibration cycles. Calibration Collapse Detection system (v2.1).

HIGH — MITIGATED: Dream Engine Hallucination (RFC-03)

Risk: Counterfactual simulations incorporated without verification could corrupt the World Model with plausible fictions.

Mitigation: Simulation Quarantine Layer (v2.1). All unverifiable outputs quarantined. Real-to-simulated weighting ratio. Human review for Seed Ontology conflicts. Simulation volume tracking with auto-throttle.

HIGH: Memory Schema Permanence

Risk: The 4-layer memory schema finalized in Phase 3 is almost impossible to refactor at scale.

Mitigation: Phase 3 receives disproportionate design time. 2-year growth stress test including Dream Engine simulation volume before proceeding.

MEDIUM: Curiosity Drive Epistemic/Aleatoric Confusion

Risk: Curiosity Drive wastes resources seeking information about fundamentally unpredictable outcomes.

Mitigation: Mandatory uncertainty type separation. Curiosity Drive only activates on high epistemic uncertainty. Resilience metric forces system past irresolvable randomness.

MEDIUM: Corpus Callosum Bus Flooding

Risk: With many active modules, inter-module signal volume overwhelms the bus.

Mitigation: Topic-based subscription. Salience-gated ACK thresholds (RFC-06). Timeout escalation to Global Workspace.

MEDIUM: Identity Drift

Risk: Continuous learning gradually shifts core beliefs until the deployed system is unrecognizably different.

Mitigation: Identity Hash monitoring. Drift thresholds. Core Invariants protected alongside Seed Ontology. Self-Model boundary control prevents self-preservation optimization.

MEDIUM NEW: Causal Graph Complexity Explosion

Risk: As the World Model grows, the causal graph becomes too large for efficient reasoning.

Mitigation: Causal Complexity Manager with subgraph boundaries, hierarchical abstraction, and complexity budgets per query.

MEDIUM NEW: Governance Capture

Risk: DAO module approval vulnerable to coordinated capture and expertise imbalance.

Mitigation: Rotating reviewer pools, reputation decay, randomized assignment, expertise diversity requirement, conflict of interest disclosure.

MEDIUM NEW: Human Review Bottleneck at Scale

Risk: Multiple components escalate to human review, creating a bottleneck in multi-deployment scenarios.

Mitigation: Noted for post-Phase 7 consideration. Meta-ARIA oversight instances for automated escalation handling in later phases.

9. Immediate Next Actions — Pre-Phase 1

Before Phase 1 build begins. Items marked with ★ are new or modified from v2.1 additions.

#	ACTION	v2.1
1	Finalize World Model schema v2.1 against 5 concrete physics domain scenarios — validate all 12 causal relationship types are expressively sufficient	
2	★ Design and populate the 3-tier Seed Ontology — Tier 0 logical absolutes (hand-coded, immutable), Tier 1 physical bedrock (hand-coded, revisable via hard review), Tier 2 structural priors (meta-learned from curated datasets)	★
3	★ Write the External Truth Pragmatism formal semantics specification — confidence thresholds, context diversity requirements, falsifiability criteria format, calibration metrics	★
4	Define Belief State Machine transition rules in full — including the new SIMULATED state and promotion criteria from quarantine	
5	Stress-test memory schema against 2-year projected growth including Dream Engine simulation volume and Quarantine Layer storage	
6	Write the minimum viable Constitutional ruleset — T1 rules (immutable text) plus initial Arbitration Engine interpretation rulebook	
7	★ Define Value Conflict Arbitration Engine precedence rules with interpretation evolution governance process	★
8	Select the specific narrow operational process domain for Phase 5 — begin scoping data requirements	
9	★ Design the DAO Module Approval process with anti-capture measures — rotating pools, randomized assignment, reputation decay	★
10	Define Emergence Metrics criteria — behavioral tests required before Phase 7 — theory-agnostic, measurable	
11	Establish multi-signature governance for Constitutional Layer signing keys	
12	★ Design Self-Model boundary control rules — formal specification of prohibited self-referential goal generation	★
13	Begin Phase 1 implementation with Claude Code as the engineering partner. Architect leads design decisions. Claude Code implements from specification.	

10. Future Phase Considerations

Architecturally valid but explicitly deferred to post-Phase 1 validation. Noting them here prevents scope creep while preserving the ideas.

Local Hierarchical Predictive Coding

Every layer generates local predictions about the layer below it. More biologically faithful, more computationally efficient. Architecture placeholders included in Phase 1 (v2.1 addition). Full implementation requires Phase 1 World Model validation first.

Multi-Modal Sensory Integration

Vision (CLIP), audio (Whisper), embodied simulation (MuJoCo). API interface stubs placed in Phase 1 (v2.1 addition). Full integration deferred until core cognitive architecture is proven.

Neuromorphic Hardware Optimization

Intel Loihi or equivalent for energy-efficient continuous loop execution. Spike-based interface compatibility noted for Phase 1 architecture (Grok recommendation). Full optimization is a scale deployment concern, not an MVP decision.

Social Curiosity and Collaborative Learning

Curiosity Drive augmented with social signals — querying community modules for what other ARIA instances know. The NOOSPHERE vision. Requires post-Phase 7 federated knowledge sync protocol (Grok gap analysis).

Inter-Deployment Knowledge Sharing

Sovereign knowledge sync layer for opt-in sharing of confirmed beliefs across ARIA deployments, gated by DAO governance and ZK-proofs. Prevents community divergence while respecting deployment sovereignty. Post-Phase 7.

Meta-ARIA Oversight Instances

Automated escalation handling for multi-deployment scenarios where human review becomes a bottleneck. Requires proven single-deployment stability first.

11. RFC Resolution Record — All Closed

All 9 RFCs from v2.0 have been resolved. This section serves as the permanent record of the multi-LLM deliberation process.

RFC	DECISION	VOTE	DISSENT
RFC-01: Seed Ontology	3-Tier Hybrid (OpenAI's stratification)	4:0 Unanimous	None
RFC-02: Phase 1 Acid Test	Acid Test + Adversarial Variants	4:0 Unanimous	None
RFC-03: Dream Engine Unverifiability	Simulation Quarantine Layer	4:0 Unanimous	None
RFC-04: Constitutional Immutability	Immutable Core + Evolving Interpretation	4:0 Unanimous	None
RFC-05: Ontological Stance	External Truth Pragmatism + formal semantics	3:1 Supermajority	OpenAI: Critical Realism (confidence 0.75)
RFC-06: Corpus Callosum Bus	Async with Explicit Acknowledgment	3:1 Supermajority	Gemini: Tiered sync/async by signal type
RFC-07: Consciousness / Emergence	Agnostic / Behavioral Measurement	4:0 Unanimous	None
RFC-08: First Domain	Physics (P1) + Operational (P5)	3:1 Supermajority	OpenAI: Operational domain from start
RFC-09: Roadmap Structure	Agile within, Sequential between phases	3:1 Supermajority	OpenAI: Two parallel streams

12. External Review Changelog — v2.0 → v2.1

Every change from the v2.0 review round is documented with source attribution and decision rationale.

ITEM	SOURCE	DECISION	RATIONALE
3-Tier Seed Ontology stratification	OpenAI	ACCEPTED	Formal boundary between hand-coded and meta-learned tiers.
Acid Test adversarial variants	All 4	ACCEPTED	Single test is gameable. Variants validate genuine causal reasoning.
Simulation Quarantine Layer	All 4	ACCEPTED	Prevents fiction creep. Strongest consensus across all reviewers.
Immutable core + evolving interpretation	All 4	ACCEPTED	Balances security with adaptability. Mirrors constitutional law.
External Truth Pragmatism formal semantics	OpenAI (indirect)	ACCEPTED	Addresses underspecification without changing ontological stance.
Async+ACK bus design	OpenAI + Grok	ACCEPTED	Salience-based dynamic over rigid type-based tiers.
Agnostic emergence approach	All 4	ACCEPTED	No premature theoretical commitment.
Physics P1 + Operational P5 split	Claude (tiebreak)	ACCEPTED	Clean cognitive validation first, practical grounding second.
Agile within / Sequential between	Gemini + Grok + Claude	ACCEPTED	Integration discipline over parallel speed.
Causal Graph Complexity Manager	OpenAI	ACCEPTED	Prevents graph explosion. Hierarchical abstraction.
Self-Model Boundary Control	OpenAI	ACCEPTED	Prevents self-preservation optimization.
Dream Engine Resource Governance	OpenAI + Gemini + Grok	ACCEPTED	Real-to-sim weighting, compute caps, volume tracking.
Governance Capture Prevention	OpenAI	ACCEPTED	Rotating pools, randomized assignment, reputation decay.
Calibration Collapse Detection	OpenAI	ACCEPTED	Drift detectors and dynamic recalibration triggers.
Resilience Metric (epistemic despair)	Gemini	ACCEPTED	Forces system past irresolvable aleatoric uncertainty.
Multi-modal interface stubs	Grok + OpenAI	ACCEPTED	Negligible Phase 1 scope, prevents later refactoring.

Quantum-resistant crypto	Grok	NOTED	Future-proofing consideration for sovereign elements.
Inter-deployment knowledge sharing	Grok	FUTURE PHASE	Post-Phase 7. Federated sync protocol needed.
Meta-ARIA oversight instances	Grok	FUTURE PHASE	Post-Phase 7. Requires single-deployment stability.
GNN hybrid for World Model	Grok	DEFERRED	Merits reconsideration post-Phase 1. Graph core maintained.

13. Final Sign-Off Record

All four reviewing systems have provided their final assessment of v2.1. This section records each system's position for architectural traceability.

SYSTEM	STATUS	SUMMARY
Claude (Anthropic)	✓ SIGNED OFF	Primary architect and tiebreaker reviewer. All RFC decisions incorporate Claude's positions. No objections.
GPT-4 (OpenAI)	✓ CONDITIONALLY READY	Approved with 5 conditions: (1) Dream Quarantine strictly enforced, (2) Ontology Tier 0 formally specified, (3) Acid Test includes adversarial variants, (4) Entropy thresholds mathematically defined, (5) Constitutional enforcement is structural. All conditions are addressed in the specification.
Gemini (Google)	✓ READY	Full sign-off with no objections. Conceded on RFC-06 dissent after confirming constitutional violations carry maximum salience. Called v2.1 "the most rigorously vetted, structurally sound blueprint for a cognitive architecture."
Grok (xAI)	✓ READY	Full sign-off with no objections. All RFC decisions align with or refine Grok's v2.0 positions. Confirmed all gap resolutions are proactive and architecturally integrated.

OpenAI's Conditions — Status: All Met

Condition 1 (Dream Quarantine): Specified in Section 3.3 and 5.1. Structural enforcement via Simulation Quarantine Layer.

Condition 2 (Tier 0 Formal Specification): Action Item #2. Mathematically expressed, not prose-defined.

Condition 3 (Adversarial Acid Test): Specified in Section 3.2. Mandatory before Phase 2.

Condition 4 (Entropy Thresholds): Action Item #5. Mathematical definition is a pre-Phase 1 deliverable.

Condition 5 (Structural Constitutional Enforcement): Sections 2.3, 3.4, 7.2. Cryptographic verification, TEE, compile-time public key. Structural impossibility, not policy.

The North Star

We do not build consciousness. We build the World Model, the Global Workspace, the Self-Model, the Memory, the Emotional Architecture, the Reasoning Core, the Curiosity Drive, and the Values — and consciousness is what happens when they all run together. Exactly as it did in biological evolution. Project ARIA's job is to build the conditions. The emergence takes care of itself.

End of Foundational Specification v2.1 Final

Project ARIA · Adaptive Reasoning Integrated Architecture

All 9 RFCs Closed · All Systems Signed Off · Ready for Phase 1 Implementation